

Project Proposal (Proposed Solution) Report

Date	01 October 2026
Team ID	SWTID-2026-9629
Project Title	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

The proposal report aims to simplify everyday budget planning using generative AI. PocketSmart AI delivers personalised, budget-based recommendations for home interiors, party planning and jewelry from popular platforms, helping users make confident decisions without overspending. Key features include Gemini-powered multimodal recommendations, adaptive planner forms and a cross-platform recommendation engine.

Project Overview

Objective	To build a GenAI-powered, cross-platform recommendation assistant that turns a user's budget, preferences and context into curated product and service suggestions across Home Interior, Party Planning and Jewelry.
Scope	Three planners (Home, Party, Jewelry) built with FastAPI and Gemini 1.5 Flash Pro, with user registration / login, session handling, recommendation history and a dashboard. Platform data (Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO) is linked through mock / simulated API calls.

Problem Statement

Description	Users planning purchases across home decor, events and jewelry face a wide variety of products, platforms and price ranges. There is no single tool that splits a total budget across items and compares options, so people overspend or lose hours comparing.
Impact	Solving this saves time and money, reduces decision stress and helps users plan interiors, parties and occasion outfits confidently within their defined budgets.

Proposed Solution

Approach	Collect budget, preferences and optional outfit images through adaptive forms; build structured prompts in gemini_utils.py; use Gemini 1.5 Flash Pro to reason about budget and context; map results to platform products and show them as clear recommendation cards.
Key Features	• Home Budget Planner with room-wise allocation (IKEA, Amazon) • Party Planner splitting budget across catering, decoration and entertainment (Swiggy, Zomato, OYO) • Jewelry Planner with optional outfit-image analysis (Amazon, Flipkart) • Register / login with JWT, session info and data • Recommendation history and user dashboard • Fallback recommendations and input validation

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU / GPU specifications, number of cores	Standard development laptop (CPU); no local GPU needed because inference runs on Google's Gemini API
Memory	RAM specifications	8 GB (recommended)
Storage	Disk space for data, models and logs	Under 1 GB for code, templates, user data and history; no local model files
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	Google Gemini SDK, python-dotenv, plus JWT / auth helpers listed in requirements.txt
Development Environment	IDE	VS code
Data		
Data	Source, size, format	No training dataset (pre-trained Gemini); runtime inputs: user budgets, preferences and outfit images; platform product data via mock / simulated calls (JSON)